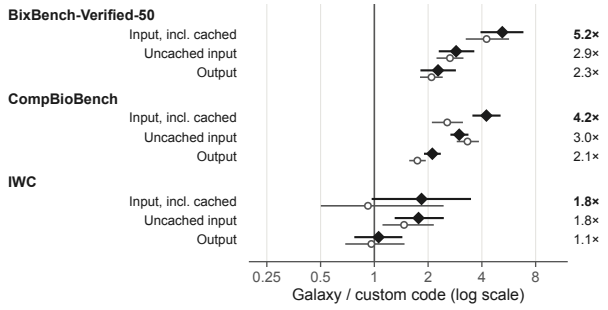
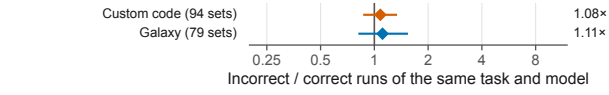
a Token use, Galaxy relative to custom code

Diamonds, typical paired task (geometric mean of task–model ratios; primary); circles, all tokens summed over the same tasks (aggregate)

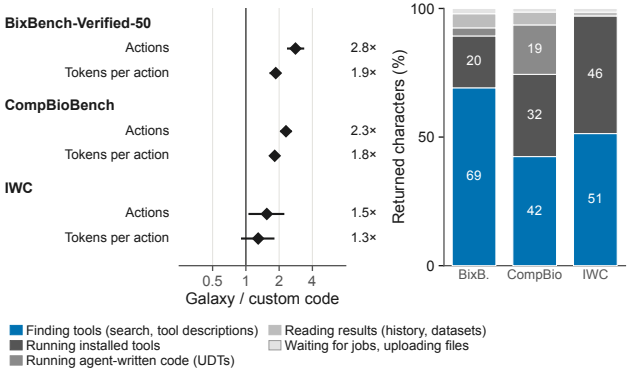


No clear input-token difference between correct and incorrect siblings



b Where the extra input comes from

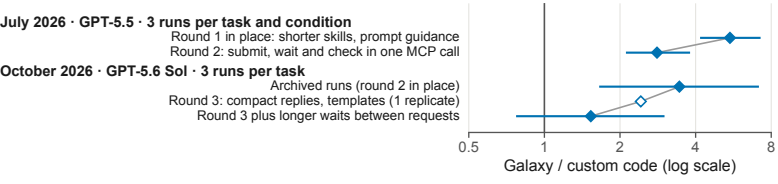
Left, Galaxy / custom code, typical paired task (all runs); right, the text Galaxy returned to the agent, by what the request was for



Tools inspected but not executed in the same run: 45–55% by benchmark.
Median share of input that was cached (earlier context reread): Galaxy 96%, custom code 93–96%.

c Reducing Galaxy token use

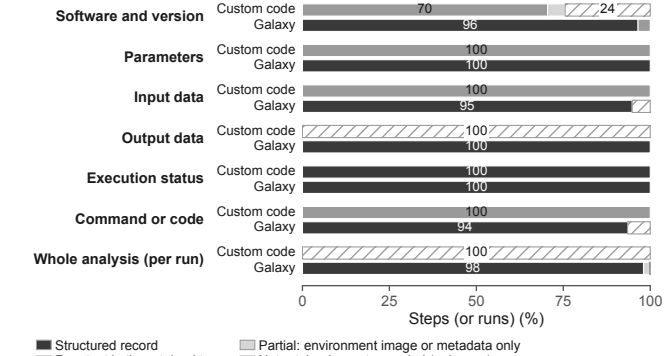
Galaxy / custom-code tokens per complete 50-task run (input, including cached, plus output) on BixBench-Verified-50, before and after each round of interface changes; filled, three replicates with 95% intervals over source capsules; open, one replicate (batch summary only). The October rows share one set of custom-code runs.



	Galaxy tokens	Custom code	Galaxy correct
July 2026 · GPT-5.5 · 3 runs per task and condition			
Round 1 in place: shorter skills, prompt guidance	5.48×	148.9M	27.2M	93%
Round 2: submit, wait and check in one MCP call	**2.81×**	66.6M (−55%)	23.7M	91%
October 2026 · GPT-5.6 Sol · 3 runs per task			
Archived runs (round 2 in place)	3.45×	122.4M	35.5M	91%
Round 3: compact replies, templates (1 replicate)	2.42×	85.9M	35.5M	–
Round 3 plus longer waits between requests	**1.53×**	54.3M (−56%)	35.5M	91%

d What the retained record holds for each analysis step

55,380 custom-code steps (shell commands labelled analysis) and 21,832 Galaxy jobs; whole analysis per run (1,908 runs per condition)



e One analysis step, recorded both ways

bix-45-q1, PhyKIT relative composition variability; GPT-5.6 Sol, replicate 1 (the case in Fig. 2d)

	Custom code: retained trace	Galaxy: job record
Software and version	PhyKIT 2.4.1 installed, then 2.0.3 downloaded and loaded through PYTHONPATH (in command text only)	phykit_metrics wrapper 0.2.0+galaxy0 (Tool Shed ID); library version not in the job record
Parameters	In the Python code of the command (file suffix, rounding)	Fields: metric = relative_composition_variability; mode = zip_archive; suffix = .mafft
Inputs and outputs	Input archives named in the command; output files not retained	1 input and 3 output datasets, linked by ID in the history
Status	Exit code 0 for every command	Job state "ok", exit code 0
Answer	Accepted	Rejected: the reference encodes the older PhyKIT value

The Galaxy record keeps fields a reviewer can query; the custom-code trace keeps the code, so the same facts must be read from it. Neither record was rerun to test reproducibility.